

DTSA 5747

Fundamentals of Natural Language Processing

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (86.8 / 100)	64.1 / 100
SPECIALIZATION Natural Language Processing		

64	90	+36
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

The field of natural language processing aims at getting computers to perform useful and interesting tasks with human language. This course introduces students to the fundamental problems in NLP, the fundamental techniques that are used to solve those problems and lays the foundation for understanding state-of-art methods. At the end of the course, students will be able to implement and analyze text classifiers, sequence labelers, discrete probabilistic models, and vector-based approaches to word meaning.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	86.8	100.0	+13.2	LOW
Applied Scientist	71.8	100.0	+28.2	MEDIUM
Machine Learning Engineer	69.4	100.0	+30.6	HIGH
Data Scientist	59.0	94.8	+35.8	HIGH
ML Engineer	58.2	77.0	+18.8	MEDIUM
MLOps Engineer	39.2	70.8	+31.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

AI Engineer

86.8 → 100.0 (+13.2 pts)

data robotics engineering learning form software technical model

Applied Scientist

71.8 → 100.0 (+28.2 pts)

lab research literature technical data learning systems industry

Machine Learning Engineer

69.4 → 100.0 (+30.6 pts)

data engineering learning tools solutions design development science

Data Scientist

59.0 → 94.8 (+35.8 pts)

data simply insights different social engineering

ML Engineer

58.2 → 77.0 (+18.8 pts)

data learning machine systems model engineering impact form

MLOps Engineer

39.2 → 70.8 (+31.6 pts)

detection data model systems learning machine image engineering

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.